MIL-C-507F
19 March 1970
SUPERSEDING
MIL-C-507E
29 April 1966

MILITARY SPECIFICATION

CLOTH, POPLIN, COTTON

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers the requirements for cotton poplin cloth.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-276	-	Thread, Cotton
PPP-P-1134	-	Packaging and Packing of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-191	-	Textile Test Methods

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8305

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2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington, D. C. 20580.)

3. REQUIREMENTS

3.1 Standard sample.- The cloth shall match the standard sample for shade and shall be equal to or better than the standard sample, with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.2 Material.- The material shall be cotton, which has been carded, combed, drawn, and spun into 2-ply yarns for the warp and single yarns for the filling.

* 3.3 Color.- The color shall be as specified and shall match the standard sample (see 6.2). The use of sulfur dyes and compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than that shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed and finished cloth shall show no more than a slight trace of labile sulfur as defined in test method specified in 4.4.

When white is specified, the cloth shall be chemically bleached and may be supplemented with optical brighteners which fluoresce to the blue region. The degree of whiteness shall match the standard sample. The finished bleached cloth shall not discolor to a degree greater than shown by the standard sample when tested as specified in 4.4. When a standard sample is not available, the discoloration shall be not less than "fair" when tested as specified in 4.4.

3.3.1 Matching.- The color shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

* 3.3.2 Colorfastness.- The dyed and finished cloth shall show fastness to light, combined laundering and bleaching, perspiration and crocking equal to or better than the standard sample when tested as specified in 4.4. When a standard sample is not available, the dyed and finished cloth shall show "good" fastness to light, combined laundering and bleaching, perspiration and no more crocking than Munsell Neutral Value 8.5 when tested as specified in 4.4.

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3.4 Physical requirements.- The finished cloth shall conform to the requirements shown in table I when tested as specified in 4.4.

TABLE I.- Physical requirements

Weight per square yard		Yarns per inch (min.)		Breaking strength (min.)		Weave
Min.	Max.	Warp	Filling	Warp	Filling	
Ounces	Ounces			Pounds	Pounds	
4.5	5.0	106	48	110	50	Plain

3.4.1 Width.- The width shall be as specified and shall be the minimum acceptable width inclusive of selvage (see 6.2).

3.5 Finish.- The cloth shall be closely singed, desized and mercerized, and shall have a clear lustrous finish.

3.5.1 Nonfibrous materials.- The starch and protein content, including chloroform-soluble and water-soluble material of the finished cloth, shall not exceed 2.0 percent when tested as specified in 4.4.

3.6 Shrinkage.- The cloth shall be preshrunk and shall not shrink more than 2.0 percent in either the direction of the warp or of the filling when tested as specified in 4.4. The preshrinking process used shall not be identified by name or trademark, either on the cloth, the ticket, or package.

3.7 Seam efficiency.- The cloth shall have a seam efficiency of not less than 85 percent when tested as specified in 4.4.

* 3.8 pH.- The pH value of the water extract of the dyed and finished cloth shall be no less than 5.5 nor more than 8.5 when tested as specified in 4.4.

3.9 Length and put-up.- Unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folds as specified in PPP-P-1134.

3.10 Fiber identification.- Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Reglations Under the Textile Fiber Products Identification Act.

* 3.11 Face identification.- The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "face" at each end of the piece

3.12 Workmanship.- The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in Section 4 shall not exceed the applicable established maximum point values.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to perscribed requirements.

* 4.1.1 Certificate of compliance.- Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.2.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents.

4.2.2 Examination of the end item.- Examination of the end item shall be in accordance with the provisions of 4.2.2.1 through 4.2.2.4.

* 4.2.2.1 Yard-by-yard examination.- Each piece in the sample shall be examined on the face side only. When the total yardage in the piece does not exceed 100 yards, the entire yardage in the piece shall be examined. When the total yardage in the piece exceeds 100 yards, only 100 yards shall be examined. All defects, as defined in Section I of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.2.1.1 except that only those slubs and knots which exceed the limits shown on figure 1 of FED-STD-4 shall be scored. No folded linear yard shall be penalized more than 4 points. The sample size shall be 20 pieces selected from 20 containers. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds the following values:

28 points for dyed cloth
32 points for white cloth

The lot shall be unacceptable if the points per 100 square yards of two or more individual pieces exceeds the following point values:

42 points for dyed cloth
48 points for white cloth

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If one piece exceeds the point level, a second sample of 20 pieces shall be examined only for individual piece quality examination. The lot shall be unacceptable if one or more pieces in the second sample exceeds the point level. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{total yards inspected}} = \text{Points per 100 square yards}$$

* 4.2.2.1.1 Demerit points.- Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension	-	one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	-	two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	-	three points
For defects exceeding 9 inches in any dimension	-	four points

NOTE: The following defects when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy or wavy cloth.
Width less than specified.
Poor dye penetration, mottled, streaky, or cloudy.
Excessive neppiness.

4.2.2.2 Examination for length.-

4.2.2.2.1 Individual pieces.- During the yard-by-yard examination, each piece shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2.2 Total yardage in sample.- The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the tickets.

* 4.2.2.3 Examination for shade.- During the yard-by-yard examination, each piece shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

* 4.2.2.4 Examination for identification of preshrinkage process, compliance with Textile Fiber Products Identification Act and face marking.- During the yard-by-yard examination, each piece in the sample shall be examined for defects listed below. The lot shall be unacceptable if two or more of the following defects are present in the sample:

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Preshrinkage process identified by name or trademark on cloth or ticket.

Not labeled or ticketed in accordance with the Rules and Regulations under the Textile Fiber Products Identification Act.

Face stamping missing from either or both ends.

Face stamping on wrong side.

* 4.3 Examination of preparation for delivery requirements.- An examination shall be made in accordance with the provisions of PPP-P-1134 to determine that packaging, packing and marking complies with the Section 5 requirement of this specification.

* 4.4 Test methods.- The methods of testing specified in FED-STD-191, whereever applicable, and as listed in table II shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The sample unit shall be 5 continuous yards full width of the finished cloth. The sample size (number of sample units) shall be as shown in the list below. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. All test reports shall contain the individual values utilized in expressing the final results.

Lot size (yards)	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and up	5

TABLE II.- Test methods

Characteristic	Requirement paragraph	Test method
Material (cotton yarn)	3.2	1200 1/
Yarn ply:		
Warp	3.2	Visual 2/
Filling	3.2	Visual 2/
Discoloration (bleached)	3.3	5660 2/ 3/
Presence of labile sulfur (dyed)	3.3	2020
Colorfastness to:		
Light	3.3.2	5660
Combined laundering and bleaching	3.3.2	5605
Perspiration	3.3.2	5680
Crocking	3.3.2	5651

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TABLE II.- Test methods (cont'd)

Characteristic	Requirement paragraph	Test method
Weight per square yard	3.4	5041
Yarns per inch:		
Warp	3.4	5050
Filling	3.4	5050
Breaking strength:		
Warp	3.4	5100
Filling	3.4	5100
Weave	3.4	Visual 2/
Identification of mercerization	3.5	Microscopic examination for swelling of cotton fiber 2/
Nonfibrous material content	3.5.1	2611
Shrinkage in laundering	3.6	5550
Seam efficiency	3.7	5110 4/
pH	3.8	2811

* 1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ One determination shall be made from each sample unit and the result reported as "pass" or "fail".

3/ Method 5660 shall be used except that the time of exposure shall be for 20 standard fading hours. Any specimen showing discoloration less than or equal to that of the standard sample shall be rated "pass". Any specimen showing discoloration greater than that of the standard sample shall be rated "fail". When a standard sample is not available, the discoloration shall be considered satisfactory if it is not worse than the rating of "fair" in the scale indicated in Method 5660.

4/ The thread shall conform to type IA1 of V-T-276. The needle size shall be 0.032 ± 0.001 inch, and the top (needle) thread shall be 60/3; the bottom (looper) thread shall be 70/2.

* 5. PREPARATION FOR DELIVERY

5.1 Put-up and packaging.- Put-up and packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C.- The cloth, put up in flat folds, shall be packaged in accordance with the applicable requirements of PPP-P-1134.

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5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Levels A, B, and C.- The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.

5.3 Marking.- In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-1134.

6. NOTES

6.1 Intended use.- The cotton poplin cloth covered by this specification is intended for use in the manufacture of clothing items.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Color of cloth required (see 3.3).
(c) Width of cloth required (see 3.4.1).
(d) Length of piece or put-up when other than specified (see 3.9).
(e) Selection of applicable levels of put-up packaging and packing (see 5.1 and 5.2).

6.3 Dyestuff formulation.-

6.3.1 Olive Green 107.- A suggested but not mandatory dyestuff formulation for Olive Green 107 is as follows:

Vat Black 25 - C.I. 69525
Vat Green 3 - C.I. 69500
Vat Green 8 - C.I. 71050

Shaded with either or both of the following:

Vat Brown 3 - C.I. 69015
Vat yellow of suitable fastness

The redness of shade should result from the predominant use of the main colors specified and not from the shading colors.

6.4 Sample.- For access to standard sample, address the procuring office issuing the invitation for bids.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army - GL
Navy - SA
Air Force - 11

Review activities:

Army - MD
Navy - SA, MC, MS
Air Force - 82

Preparing activity:

Army - GL

Project No. 8305-0627

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Cloth, Poplin, Cotton MIL-C-507F

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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